Teacher-in-the-Loop AI Feedback for English Language Learning: A Pilot Study with ESL/English Teachers at Bartar Language School

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Executive Summary

This report presents a pilot study in which eight English/ESL teachers at Bartar Language School reviewed four AI-assisted writing feedback prototypes. Bartar Language School provides English-language education to learners whose first language is not English. The study explored how AI feedback should be designed so that teachers remain in control, students continue to revise actively, and feedback does not become a shortcut for copying.

Across the results, teachers strongly preferred designs that kept AI behind the teacher or delayed full corrections until after student effort. Prototype C, Teacher-Controlled Feedback, was selected as the best overall design by four of eight teachers, while Prototype D, Staged Feedback, was selected by three teachers. Prototype A, Direct Correction, was viewed as clear and fast, but all eight teachers identified it as the highest copying-risk design.

- Teachers wanted AI to act as a drafting and diagnostic assistant, not as an autonomous feedback authority.
- The strongest teacher-control requirements were approval before student visibility, editing rights, the ability to hide the full answer, and the ability to require a student revision attempt first.
- Teachers saw productive value in AI hints and mini-examples, but preferred that model answers be staged, delayed, or teacher-approved.
- The final design implication is a Teacher-in-the-Loop Feedback Control Framework built around teacher approval, hint-first feedback, student revision, uncertainty visibility, and optional teacher notes.

Pilot indicator	Scenario result
Participants	8 Bartar English/ESL teachers
Most preferred prototype	Prototype C - Teacher-Controlled Feedback (4 of 8)
Second strongest design	Prototype D - Staged Feedback (3 of 8)
Highest copying-risk prototype	Prototype A - Direct Correction (8 of 8)
Top requested feature	Teacher approval before student visibility (8 of 8)

1. Context and Rationale

Generative AI tools can now produce grammar corrections, rewritten sentences, vocabulary suggestions, and explanations within seconds. For English language learning, this creates a double-edged design problem. Fast AI feedback may help students receive more practice, but answer-giving feedback may also reduce student effort, encourage copying, and make it harder for teachers to judge what students actually understand.

The project therefore did not ask teachers a general question such as, "What do you think about AI?" Instead, it asked a design question: how should AI feedback be structured so that teachers remain in control and students still learn from revision? This framing aligns with human-in-the-loop approaches to educational AI. The U.S. Department of Education has emphasized that educators need the ability to exercise professional judgment and override AI models when necessary, and UNESCO frames AI teacher competency around human agency, ethics, pedagogy, and professional learning.

The setting for the scenario is Bartar Language School, an English-language education context where teachers regularly support learners whose first language is not English. This makes the feedback-design problem practical: teachers need feedback that is efficient, level-appropriate, and supportive, while still preserving student voice and learning responsibility.

2. Research Questions

- RQ1. How do English/ESL teachers evaluate different AI feedback designs in terms of learning support, copying risk, trust, teacher control, and classroom usability?
- RQ2. Which teacher-control features are considered necessary before AI-generated feedback is shown to students?
- RQ3. How can AI feedback be redesigned to preserve student revision effort rather than simply provide answers?

3. Methodology

The study was designed as an exploratory mixed-method co-design study. Teachers reviewed four hard-coded AI feedback prototypes through a web prototype. During review, the researcher used a think-aloud protocol to capture immediate reactions. Teachers then completed a prototype comparison survey and participated in a short co-design discussion.

Component	Description
Number of teachers	8
Institution	Bartar Language School
Teaching context	English/ESL instruction for learners whose first language is not English
Data collection activities	Background survey, prototype review, think-aloud interview, comparison survey, co-design discussion
Student data	No real student data used; synthetic writing sample only

3.1 Prototype Conditions

Prototype	Design	Purpose
A - Direct Correction	AI gives the full corrected paragraph immediately.	Tests speed and clarity versus copying risk.
B - Guided Hint	AI gives hints and revision prompts without full correction.	Tests whether hints preserve student thinking.
C - Teacher-Controlled	AI suggests categories and feedback; teacher chooses what students see.	Tests teacher agency, visibility control, and trust.
D - Staged Feedback	Student receives hints first and stronger correction after attempting revision.	Tests student effort preservation and staged access.

3.2 Measures and Analysis

- Quantitative ratings used 1-5 scales for learning support, copying risk, teacher control, student effort, trust, and likelihood of classroom use.
- Ranking questions identified the best overall design, highest copying-risk design, and most teacher-controlled design.
- Think-aloud and interview notes were coded using a thematic codebook covering teacher control, student effort, copying risk, trust, workflow, student level fit, and teacher editing behavior.
- Co-design notes were converted into design requirements for a final Teacher-in-the-Loop Feedback Control Framework.

4. Results

4.1 Prototype Ratings

The ratings show a clear pattern. Direct Correction scored highest for speed and clarity in teacher comments, but its overall use score was low because teachers associated it with answer-copying. Teacher-Controlled Feedback and Staged Feedback scored highest on teacher control, student effort, trust, and likely classroom use.

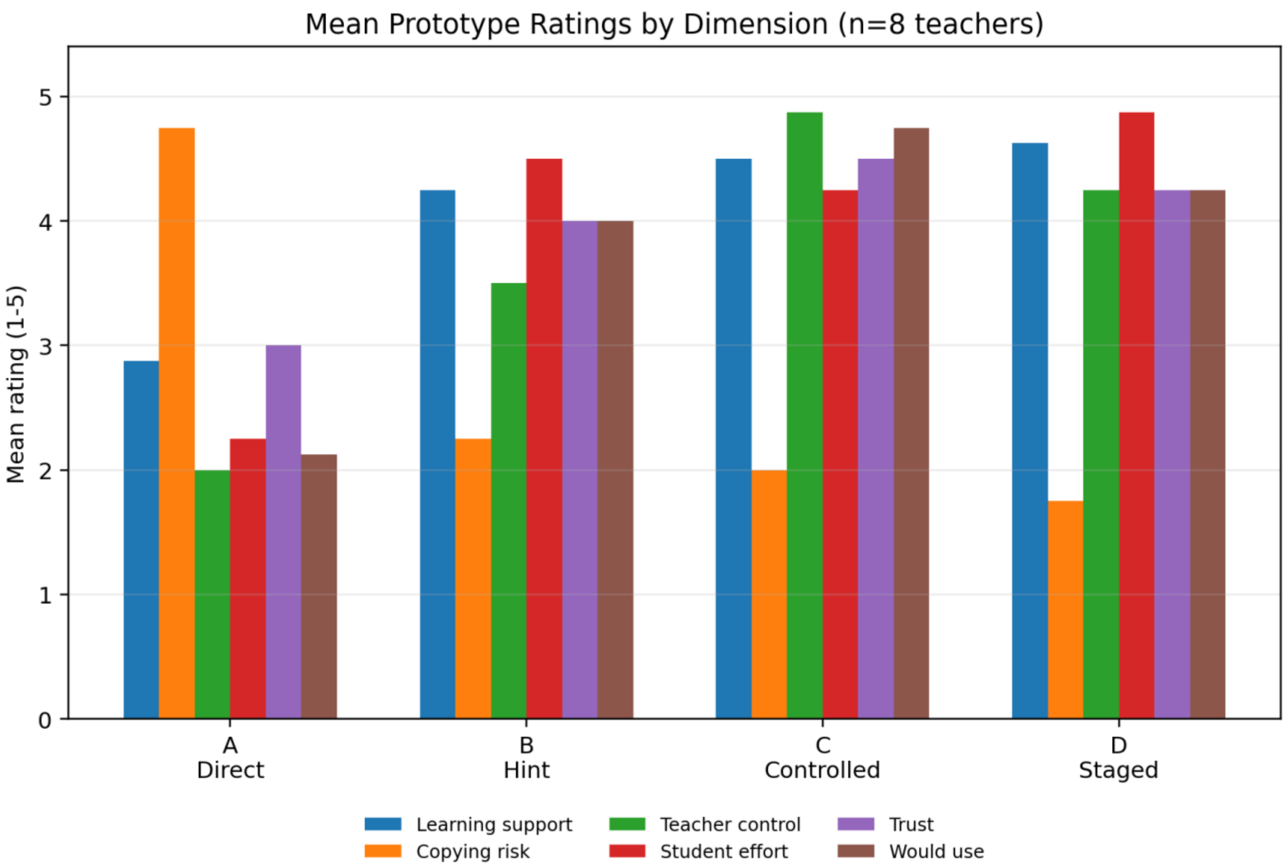

Figure 1. Mean prototype ratings across six evaluation dimensions (n=8).

Prototype	Learning support	Copying risk	Teacher control	Student effort	Trust	Would use
A - Direct Correction	2.88	4.75	2.00	2.25	3.00	2.12
B - Guided Hint	4.25	2.25	3.50	4.50	4.00	4.00
C - Teacher-Controlled	4.50	2.00	4.88	4.25	4.50	4.75
D - Staged Feedback	4.62	1.75	4.25	4.88	4.25	4.25

4.2 Overall Prototype Preference

Prototype C was selected as the best overall design by half of the teachers. Prototype D was close behind, suggesting that teachers value both teacher control and staged student effort. Prototype B was seen as useful but less complete because it did not give teachers as much control as Prototype C. Prototype A was not selected as best overall by any teacher.

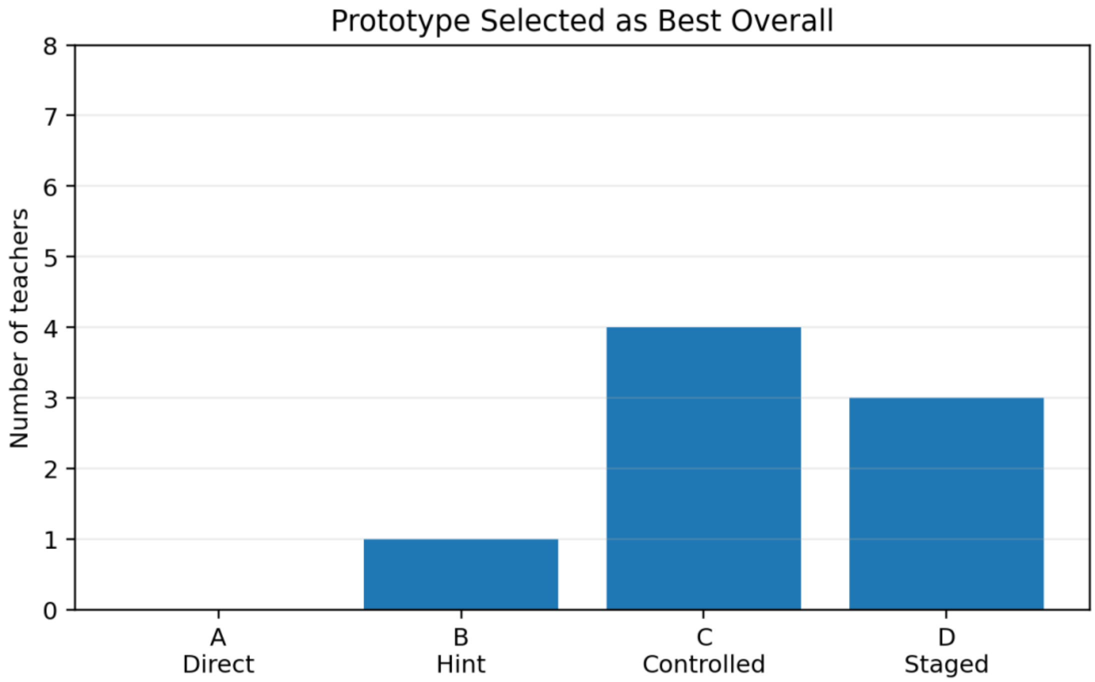

Figure 2. Prototype selected as best overall design (n=8).

4.3 Copying Risk and Student Effort

All eight teachers selected Prototype A as the highest copying-risk design. In interviews, teachers described full correction as efficient but pedagogically risky because students could submit the corrected paragraph without processing the underlying grammar pattern. Prototype D received the lowest copying-risk rating because it required a revision attempt before showing stronger correction.

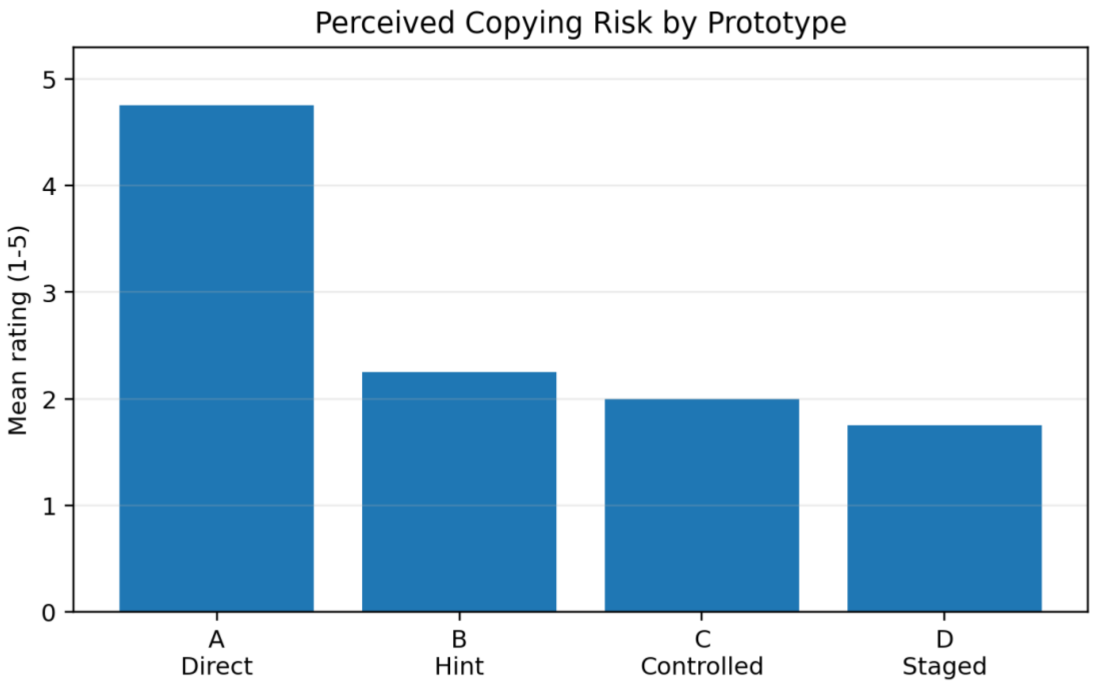

Figure 3. Perceived copying risk by prototype (n=8).

4.4 Teacher-Control Requirements

The strongest design requirement was teacher approval before student visibility. All eight teachers selected teacher approval and teacher editing as essential. Seven teachers also selected the ability to hide the full answer and require a student revision attempt before revealing model feedback.

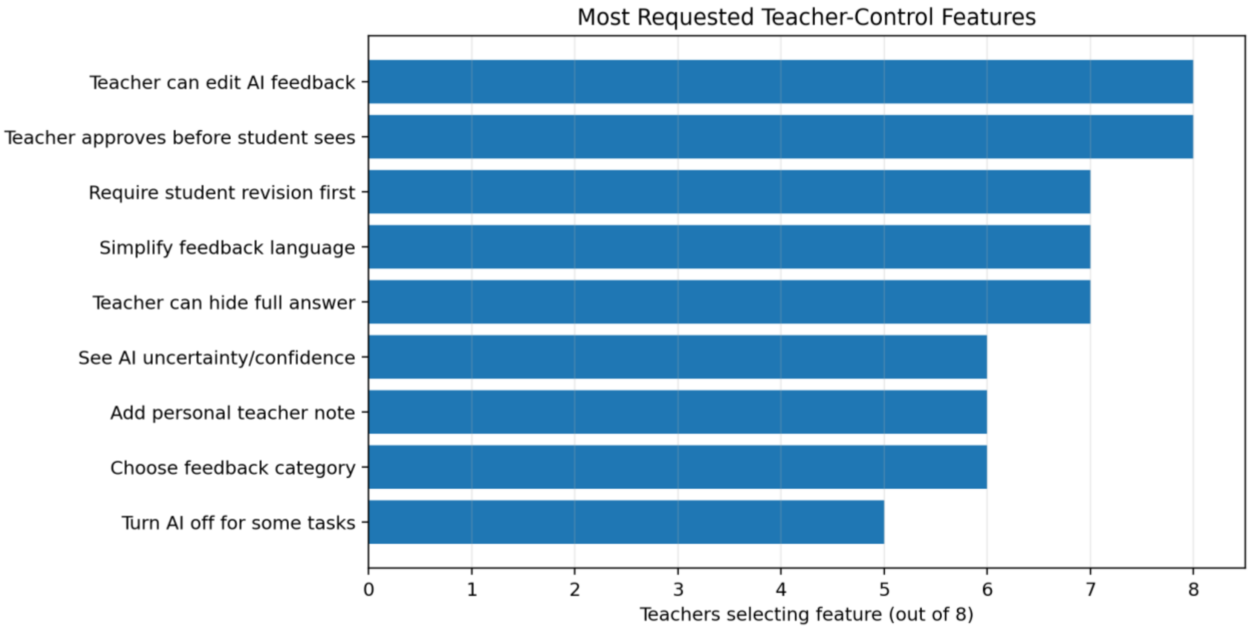

Figure 4. Most requested teacher-control features (n=8).

4.5 Qualitative Themes

Thematic coding of think-aloud notes and interviews produced eight major themes. The most frequent theme was teacher control, followed by student effort and copying/over-reliance. Workflow also appeared frequently: teachers did not want AI feedback to create a second marking burden.

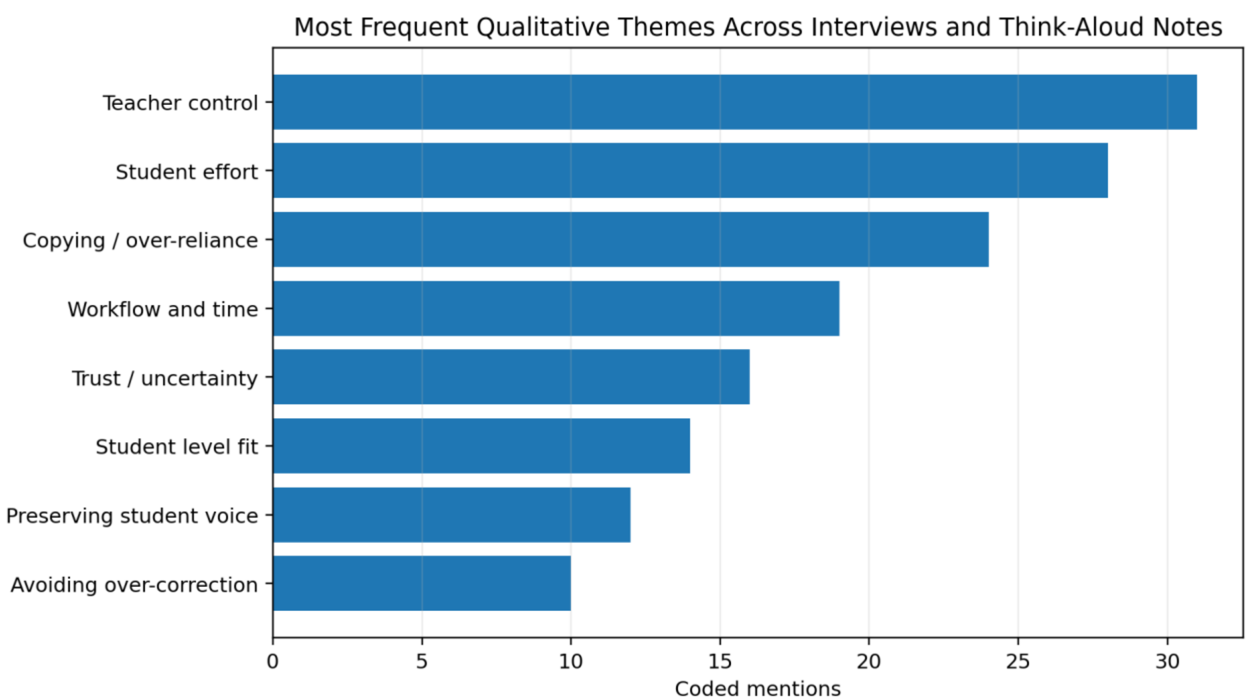

Figure 5. Thematic analysis counts from think-aloud and interview notes.

5. Qualitative Findings

Theme 1: Teachers want AI as an assistant, not an authority

Teachers repeatedly distinguished between AI as a helpful assistant and AI as a replacement for teacher judgment. The teacher-controlled prototype was preferred because it allowed AI to identify possible issues while leaving decisions about tone, level, visibility, and feedback depth to the teacher.

T02: "I would use AI to prepare options, but I would not want it to speak to my students without me checking it first."
T05: "The teacher knows whether this student needs grammar correction today or just encouragement to keep writing."
T07: "The AI can notice patterns, but the teacher should decide what matters most."

Theme 2: Direct correction is fast but weak for learning

Teachers acknowledged that Prototype A could save time, especially for simple grammar errors. However, most argued that direct correction should be reserved for review, not first feedback. The main concern was that students might copy the corrected version without understanding why changes were made.

- Teachers wanted full correction hidden by default.
- Teachers preferred hints, examples, or error categories before full model answers.
- Teachers saw full correction as useful after revision, not before revision.

Theme 3: Staged feedback protects student effort

Prototype D was valued because it made revision effort visible. Teachers liked that students first received hints, then attempted revision, then compared their answer with a model. This staged structure was described as slower but more pedagogically defensible.

Theme 4: Feedback must fit student level and teacher voice

Teachers were concerned that AI explanations could be too advanced, too long, or too confident. They wanted controls to simplify feedback language, soften the tone, add encouragement, and avoid changing the student’s voice. Teachers also emphasized that not every error should be corrected at once.

6. Co-Designed Feedback Control Framework

Based on the teacher responses, the final recommended artifact is a Teacher-in-the-Loop Feedback Control Framework. The framework is organized around five design principles.

Design principle	Implementation meaning
1. Teacher approval before visibility	AI-generated feedback should be shown to the teacher first. The teacher decides what the student sees.
2. Hint before answer	Students should usually see hints, categories, or mini-examples before full corrected sentences.
3. Student revision before model correction	The system should require or encourage an attempt before revealing stronger feedback.
4. Teacher-editable tone and level	Teachers must be able to simplify language, change tone, hide parts, and add a personal note.
5. AI uncertainty visible to teachers	Confidence or uncertainty information should be available to teachers, but not necessarily shown to students.

6.1 Recommended Student-Facing Feedback Template

Step	Element	Example wording
Step 1	Encouragement	You explained your daily routine clearly. Good work.
Step 2	One main issue	Today, focus on simple present verbs for daily routines.
Step 3	Hint	For daily habits, use the base verb: I prepare, I go, I watch.
Step 4	Student revision prompt	Try to revise these sentences yourself first.
Step 5	Mini-example after attempt	I preparing breakfast -> I prepare breakfast.
Step 6	Teacher note	Focus on verb forms first. Do not worry about making the paragraph perfect today.

7. Implications

7.1 Implications for Bartar Language School

- Use AI feedback as a teacher-support tool first, not as a direct student-facing correction tool.
- Adopt a hint-first policy for student writing feedback, especially for homework and revision tasks.
- Train teachers to review, edit, and simplify AI-generated feedback before classroom use.
- Create a shared feedback bank for common ESL issues such as simple present, subject-verb agreement, uncountable nouns, and translation-style phrasing.
- Avoid AI detection-style accusations; instead, respond to AI-like writing through voice-preservation and student explanation tasks.

7.2 Implications for Tool Design

- Default setting should be teacher-first visibility.
- The interface should include controls for show hint only, hide answer, require revision, simplify language, and add teacher note.
- Full corrected paragraphs should be available, but not shown automatically.
- The tool should produce a student revision prompt, not only a correction.
- AI uncertainty should be framed as teacher information, not student confusion.

7.3 Implications for Student Learning

The study suggests that AI feedback becomes more educationally useful when it preserves friction in the revision process. Productive friction means the student has to notice, attempt, compare, and revise rather than immediately receive a polished answer. In ESL writing, this is especially important because the student's developing voice and error patterns are part of the learning evidence.

8. Limitations

- The sample size of eight teachers is appropriate for an exploratory design study, but not for broad statistical generalization.
- Only one synthetic writing sample was used in the prototype comparison; future testing should include translation-style writing, organization problems, academic tone issues, and AI-like writing.
- Student learning outcomes were not directly measured in this teacher-only pilot.

9. Recommended Next Steps

Next step	Action
Step 1	Revise Prototype C and D into a combined teacher-controlled staged feedback design.
Step 2	Pilot the revised design with 10-15 students using synthetic or anonymized writing tasks.
Step 3	Compare student revisions before and after hint-first feedback.
Step 4	Develop a short teacher guide for responsible AI feedback use at Bartar.

10. Conclusion

The Bartar pilot indicates that teachers are not simply for or against AI feedback. Their acceptance depends on design. Teachers were most supportive when AI helped them identify patterns, draft feedback, and save time while preserving teacher judgment and student effort. The strongest design direction is therefore not a fully automated AI corrector, but a teacher-controlled, staged feedback assistant.

The central design recommendation is to combine Prototype C and Prototype D: AI should first support the teacher, the teacher should decide what students see, and students should revise before receiving full corrections. This model aligns with responsible AI principles because it keeps teachers in the loop and keeps students cognitively active.

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Appendix A. Prototype Comparison Survey Dimensions

Dimension	Survey item
Learning support	This feedback would help students learn.
Copying risk	This feedback may encourage students to copy instead of learn.
Teacher control	This feedback gives the teacher enough control.
Student effort	This feedback supports student revision effort.
Trust	I trust this feedback style.
Would use	I would use this in my classroom.

Appendix B. Qualitative Codebook

Code	Theme	Definition
TC	Teacher control	Comments about approval, editing, visibility, override, and teacher judgment.
SE	Student effort	Comments about revision, thinking, attempting before correction, and productive struggle.
CR	Copying risk	Comments about direct answer copying or over-reliance on AI.
TR	Trust	Comments about confidence, uncertainty, reliability, and teacher willingness to use the feedback.
WF	Workflow fit	Comments about time, classroom practicality, and teacher workload.
TL	Student level fit	Comments about feedback matching learner level and language ability.
VOICE	Preserving student voice	Comments about not rewriting the student into an unnatural or overly advanced style.